

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (CL)

April/May 2012

CL 201 Surveying

Date: 01.05.2012, Tuesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Describe any four various accessories required for plane table surveying. [04]
(b) Explain the procedure of setting up plane table over a station. [03]
- Q - 2 (a) Following are the observations for closed traverse. Calculate the latitude and [05]
Departures. Also calculate closing error for the traverse using Transit Rule.

Line	Length	R.B
AB	89.30	45°20'NE
BC	210.50	70°10'NE
CD	150.40	180°10'SE
DE	160.10	43°00'SW
EA	232.30	58°20'NW

- (b) Enlist methods of Temporary Adjustment of a Theodolite. Describe Any Two in [04]
details.
- (c) Show with neat sketches the characteristic features of contour lines of the following. [05]
(1) A pond (2) A hill (3) A ridge line (4) A valley (5) A vertical cliff

OR

- Q - 2 (a) The following records are obtained in a traverse survey, where the length and bearing [05]
of the last line were not recorded. Compute the length and bearing of line DA.

Line	Length	Bearing
AB	75.50	30°24'
BC	180.50	110°36'
CD	60.25	210°30'
DA	(?)	(?)

- (b) Discuss Repetition Method of horizontal angle measurement using Theodolite. [04]

- (c) Explain sources of error in Levelling. [05]
- Q - 3 (a) Enlist the various rules to compute the area of a traverse. [02]
- (b) A railway embankment is 12 m wide. The ground is level in a direction transverse to the centre line. Calculate the volume contained in a 100 m length by trapezoidal rule and prismoidal rule, if the side slope is 1.5:1. The centre heights at 20 m interval are 3.7 m, 2.6 m, 4.0 m, 3.4 m, 2.8 m, 3.0 m, 2.2 m. [05]
- (c) Enlist types of horizontal and vertical curves. [03]
- (d) Define Transition curve. Explain requirements of ideal transition curve. [04]

OR

- Q - 3 (a) Enlist the various rules to compute the volume of traverse. [02]
- (b) Determine the area between the line AB and a meandering stream for offsets taken at a regular interval of 20 m along the line AB by Trapezoidal rule and Simpson's rule. [05]

Point	A								B
Distance (m)	0	20	40	60	80	100	120	140	160
Offset length (m)	23	40	42	30	32	60	10	14	22

- (c) Calculate the length of a transition curve, when rate of radial acceleration is 30 cm/s^3 , allowable speed on curve is 60 km/h and the radius of the circular curve is 200 m. [03]
- (d) Draw a neat sketch of elements of a simple circular curve. [04]

SECTION - II

- Q - 4 Describe the method of setting out of Buildings. [07]
- Q - 5 (a) Describe the method of setting out of culverts. [09]
- (b) Define the term sounding. Enlist the equipment used for sounding. [05]

OR

- Q - 5 (a) Describe the procedure of setting out position of bridge pier. [09]
- (b) Explain the different methods of locating sounding positions. [05]
- Q - 6 (a) Explain the basic features of a total station. [07]
- (b) Write a short note on Electronic Theodolite. [07]

OR

- Q - 6 (a) Explain the functioning and capabilities of a Total Station. [07]
- (b) Write a short note on any one EDM instrument. [07]
